



EXECUTIVE MBA | IT STRATEGY

Chapter 5 - Group Presentation

Data Resource Management

Managing data as a strategic business asset

From the logical data hierarchy and database structures to data warehousing, mining, and the DBMS.

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1 Agenda & Learning Objectives

What this chapter teaches and how today's session is organized.

Learning Objectives

- Explain the business value of data resource management
- Compare the database approach with traditional file processing
- Show how a DBMS supports business operations and management
- Describe the major types of databases
- Explain data warehouses and data mining
- Outline logical data elements and database structures
- Describe how databases are developed

Today's Roadmap



Foundations

Why data matters + the logical data hierarchy



Approaches & the DBMS

File processing vs the database approach



Structures

Hierarchical, network, relational, OLAP, object



Types of databases

Operational, distributed, external, warehouses



Using the data

Data mining, interrogation (SQL), a real case

2 Data as a Strategic Resource

Why data resource management matters

Data resource management is a managerial activity that applies information systems technologies, such as database management and data warehousing, to manage data as a valuable organizational resource that meets the information needs of business stakeholders.



Data is an asset

Well-managed data is a competitive resource, not just a by-product of operations.



Better decisions

Clean, integrated data feeds analytics, reporting, and decision support.



One version of truth

Shared databases remove redundancy and conflicting records across the business.



Control & trust

Managed data improves security, integrity, privacy, and regulatory compliance.

3 Logical Data Elements

How data is logically organized, from smallest to largest



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Two Approaches to Managing Data

Traditional file processing vs the database approach

Traditional File Processing

Each application keeps its own independent files of specialized data records.

- Data redundancy - the same data is duplicated in many separate files
- Lack of data integration - data is trapped in isolated, application-specific files
- Data dependence - programs are tied to file formats and storage details
- Lack of integrity / standardization - no consistent definitions or quality control

Database Management Approach

Data is consolidated into shared databases accessed by many applications through a DBMS.

- Minimal redundancy - data is stored once and shared across applications
- Integrated data - logically related data is connected in one place
- Data independence - the DBMS separates programs from physical storage
- Integrity & security - central control of quality, access, and standards

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The Database Management System (DBMS)

The software interface between users and databases

A DBMS consolidates records into databases that many applications can share. On mainframe and server systems it is used to create databases and applications, maintain data quality, and use the data to deliver information to end users.



Database Development

Define and organize the content, relationships, and structure of the data (the schema).



Application Development

Build queries, forms, reports, and web pages using the DBMS tools.



Database Interrogation

Let end users query the data and generate reports on demand.



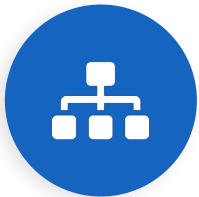
Database Maintenance

Add, delete, update, and correct data to reflect new transactions and events.

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Fundamental Database Structures

Five ways data elements can be organized and related



Hierarchical

Tree of records; one-to-many parent-child links.



Network

Records linked in many-to-many relationships.



Relational

Data in tables of rows and columns; the most widely used.



Multidimensional

Data viewed as cubes; powers OLAP analytics.



Object-Oriented

Data and the methods that act on it bundled as objects.

The next two slides look at each structure, then we evaluate their strengths and limits.

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Structures in Depth (1 of 2)

Hierarchical, network, and relational

Hierarchical

- Earliest DBMS structure
- Records arranged in a tree
- Relationships are one-to-many
- Best for structured, routine transactions

Network

- Used in some mainframe DBMS packages
- Supports many-to-many relationships
- More flexible than hierarchical
- Hard to handle ad hoc requests

Relational

- The most widely used structure
- Data stored in tables (rows = records, columns = fields)
- Links files that share a common data element
- Responds easily to ad hoc queries

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Structures in Depth (2 of 2)

Multidimensional and object-oriented



Multidimensional

- A variation of the relational model
- Organizes data into multidimensional cubes
- Each cell holds a value across several dimensions
- Popular for analytical databases and OLAP
- Fast slice-and-dice analysis for managers



Object-Oriented

- Bundles data with the methods that act on it (objects)
- Supports complex data types: graphics, audio, video
- Inheritance lets new object types reuse existing ones
- Used in multimedia and web-based applications
- Foundation for object-relational databases

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Evaluating Database Structures

Strengths and limitations at a glance

Structure	Strengths	Limitations
Hierarchical	+ Works for structured, routine transactions	- Cannot handle many-to-many relationships
Network	+ More flexible than hierarchical	- Struggles with ad hoc requests
Relational	+ Handles ad hoc queries; easy to work with and maintain	- Less fast/efficient than hierarchical or network
Multidimensional	+ Excellent for OLAP and fast analytics	- Specialized; built for analysis, not transactions

10 Database Development

How databases are designed and documented



Database Administrator (DBA)

Leads enterprise-wide database development and safeguards integrity and security.



Data Definition Language (DDL)

Used by the DBA to define data content, relationships, and structure (the schema).



Data Dictionary

A repository of metadata - data about the data: definitions, security, and usage rules.



Entity Relationship Diagram

Models entities and how they relate before the database is built.

Logical view = how end users see the data; **Physical view** = how the data is actually stored on hardware. The DBMS bridges the two.

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Types of Databases

The major categories found across an organization



Operational

Detailed data that runs day-to-day operations (customer, inventory, HR).



Distributed

Copies or parts of databases held on servers in multiple locations.



External

Data from the web or paid online services (search engines, statistics).



Hypermedia

Hyperlinked pages of multimedia rather than interrelated data records.



Data Warehouse

Cleaned, integrated, historical data used for analysis and decisions.

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Operational & Distributed Databases

Running the business, across many locations

Operational Databases

- Store detailed data that supports daily operations
- Also called subject area, transaction, or production databases
- Examples: customer, inventory, and human-resource databases

Distributed Databases

- Copies or parts of databases on servers in many locations
- Pros: local control, better performance, data protection
- Cons: keeping every copy accurate and consistent is hard



Replication

Find changes in each distributed copy and apply them to all the others. Very complex; needs extra computing power and bandwidth, but stays current.



Duplication

Treat one database as the master and copy it to all locations after hours. Much easier to accomplish, but copies lag the master.

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External & Hypermedia Databases

Data beyond the organization's own systems



External Databases

- Available on the web, often for a fee or via online services
- Search engines such as Google index huge external databases
- Include statistical, bibliographic, and full-text databases
- Give the business access to data it does not own or produce



Hypermedia Databases

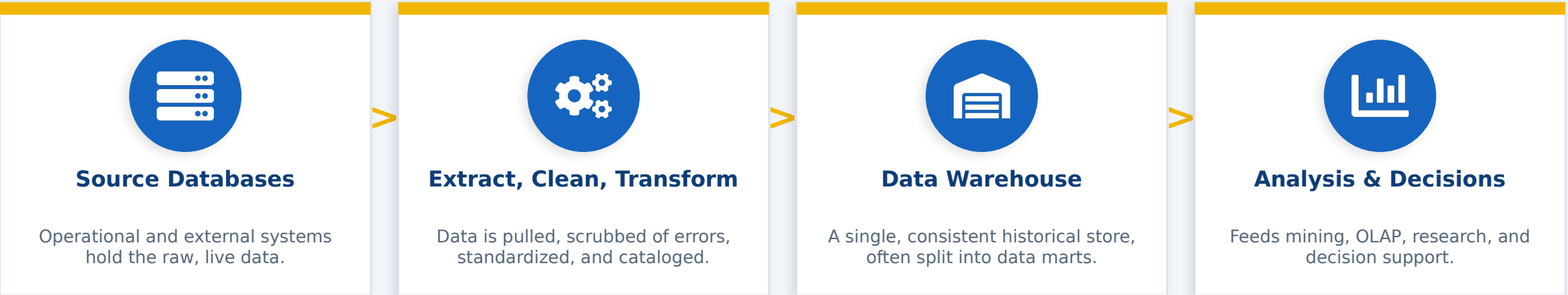
- Power web-based information systems
- Hold hyperlinked pages of multimedia content
- Interrelate hypermedia page elements, not data records
- Underlie websites, intranets, and extranets

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Data Warehouses & Data Marts

A clean, central store of historical data for analysis

A data warehouse is a central source of data that has been extracted from other databases, then cleaned, transformed, and cataloged. It stores static, historical data for analysis, research, and decision support. Subsets focused on one department or process are called data marts.



In practice: Applebee's analyzes daily sales to decide what to reorder and promote, and Travelocity mined 600,000 customer comments to monitor and respond to issues faster.

15 Data Mining

Turning warehoused data into insight

Data mining analyzes the large pools of data in a data warehouse to discover hidden patterns, trends, and relationships that support better business decisions.



Find patterns

Reveals correlations and trends humans would miss in raw data.



Market-basket analysis

Shows which products customers tend to buy together.



Segment & target

Groups customers to personalize offers and marketing.



A caution

Past-data focus can constrain thinking; pair it with fresh ideas.



SQL - Structured Query Language

The international standard query language built into most DBMS packages.

```
SELECT name, salary
FROM employees
WHERE dept = 'Sales';
```

Query form: SELECT (fields) ... FROM (tables) ... WHERE (conditions).

Report generators let users format results as reports with no programming.



Boolean Logic

Three operators - AND, OR, NOT - refine a search to exactly the data you want.

Example: cats OR felines AND NOT dogs



Graphical & Natural Queries

- Point-and-click tools translate clicks into SQL
- Natural-language queries read like conversational English
- Both spare users from hand-writing tricky SQL

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Real-World Case: Opening the Data Vault

Amazon, eBay & Google as data resources

When data is a managed resource, it can become a platform. These firms opened parts of their databases to outside developers - turning data into ecosystems and revenue.



Amazon

Spent ~10 years and \$1bn building its data vault (products, prices, rankings, reviews, inventory). Opened it in 2002; 65,000+ developers and businesses tapped in, many becoming partners.



eBay

Opened its \$3bn databases in 2003. 15,000+ developers registered, 1,000+ new applications appeared, and 41% of listings were uploaded using these resources.



Google

Moved cautiously: free access to 1,000 search results a day, more by permission. Opened AdWords to outside applications in 2005, spawning dozens of services.

Discussion: should every company open its databases to outsiders? Weigh the new revenue and innovation against losing control of a core asset.

18 Key Takeaways

What to remember from Chapter 5



Manage data as a resource

Data resource management treats data as a strategic asset that serves stakeholder information needs.



Database approach wins

Shared databases beat file processing by cutting redundancy and adding integration, independence, and integrity.



Relational dominates

Of the five structures, the relational model is the most widely used; multidimensional powers OLAP analytics.



Warehouse then mine

Cleaned, historical data in warehouses fuels data mining and decision support.



The DBMS is the engine

It develops, maintains, and interrogates databases, with SQL as the standard query language.



Govern the data

DBAs, data dictionaries, and security keep enterprise data accurate, consistent, and protected.



Thank You

Questions & Discussion

Chapter 5 - Data Resource Management

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